Aaron Alphons Thomas

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EDUCATION

Vellore Institute of Technology(VIT) Vellore, India
Integrated (Mtech) – Software Engineering; CGPA:9.39 *Aug 2023- Present*
Courses: Operating Systems, Data Structures, Computer Networks, Database System Management

SKILLS SUMMARY

- **Languages**: Java, SQL, Python, C, C++, Html, CSS, Javascript
- **Framework**: VS Code
- **Tools & Platforms**: Git
- **Soft Skills**: Communication, Time Management, Problem-Solving, Team Collaboration

EXPERIENCE

Persistent Systems *May 2025- Present*

AI/ML Intern (Remote)

- **Smart Traffic Monitoring System** — *Flask, Python, YOLOv8, OpenCV, Google Maps API, JavaScript/HTML/CSS.* Detect and track multiple vehicle types with over 95% accuracy, along with lane detection and lane-change alerts.
- Developed AI-driven traffic analytics including live speed estimation, traffic-density classification (Low/Moderate/High/Severe), type-wise vehicle counting, and congestion-zone visualization with GPS-based routing and satellite map overlays through Google Maps APIs.
- Designed a responsive web dashboard with real-time charts, analytics panels, CSV/PDF export for traffic reports, and voice-enabled features such as dashboard voice control and automatic weather announcements.

EXTRACURRICULAR

Riviera – VIT

Served as a Coordinator for Riviera, the 4-day annual cultural fest at Vellore Institute of Technology, managing event coordination, participant support, and on-ground operations.

Larson and Tourbo (L&T)

Assisted in an NDA signed-project. Worked in Technical Team designing the logic and the workflow of the code.

PROJECTS

Blood Donation Website — *HTML, CSS, JavaScript, PHP*

- Developed a web platform to connect blood donors and recipients, enabling quick search and real-time matching based on blood group, location, and availability.
- Implemented secure user authentication with registration and login features to manage donor and recipient profiles.

Smart Pantry Management System — *IoT, Embedded Systems*

- Developed an IoT-based Smart Pantry system using RFID, ultrasonic, and vibration sensors for real-time food inventory tracking and accurate detection of item placement or removal.
- Built an OLED-based live inventory display and inventory-driven dish recommendation feature on a web dashboard using available ingredients.

Ransomware Simulator — *Network Security*

- Developed a ransomware simulation tool that detects corrupted files and demonstrates file encryption and decryption workflows for security testing and analysis.
- Implemented automated corrupted-file detection with configurable auto-deletion of infected files to simulate basic incident response and recovery behavior.

LICENSES AND CERTIFICATES

Project Management Certificate by Google
AWS Cloud Technical Certificate

HACKATHON AND ACHIEVEMENTS

Cyber-verse Hackathon — Vellore Institute of Technology, Vellore (36-hour Hackathon) Runner-Up
Impact-AI Hackathon — St. Josephs College Of Engineering, Chennai (48-hour hackathon) Winner (AI/ML)